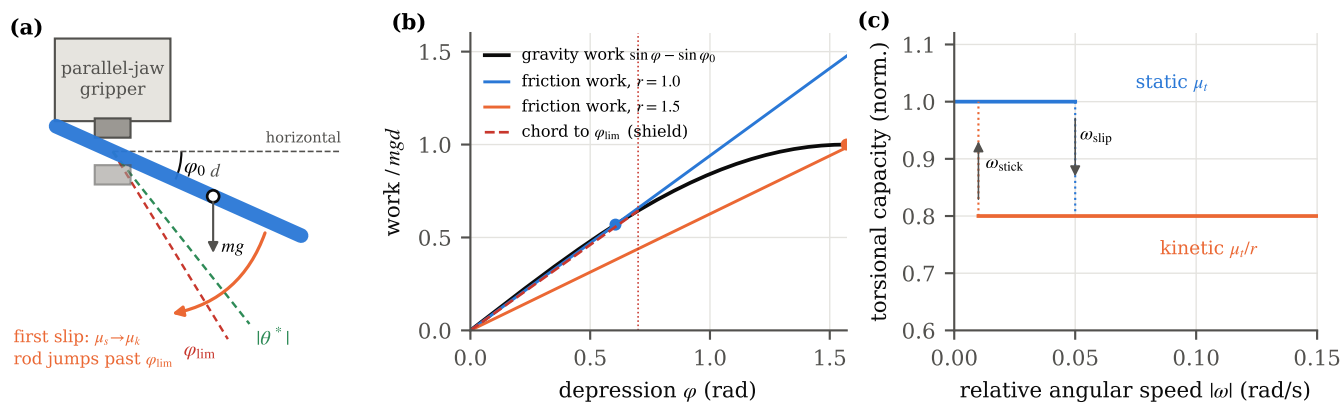
(a)
parallel-jaw gripper
horizontal
φ_0 d
mg
first slip: $\mu_s \to \mu_k$
rod jumps past φ_{lim}
φ_{lim}
$|\theta^*|$
(b)
gravity work $\sin\varphi - \sin\varphi_0$
friction work, $r = 1.0$
friction work, $r = 1.5$
chord to φ_{lim} (shield)
work / mgd
depression φ (rad)
(c)
static μ_f
kinetic μ_f/r
ω_{stick}
ω_{slip}
torsional capacity (norm.)
relative angular speed $|\omega|$ (rad/s)